PDF OPERATION MANUAL

VCA Synthetic Customer Lab

How a Visa/VCA reviewer can upload a real survey/interview PDF as the scenario artifact, run Swiss synthetic customer agents with IBM watsonx.ai / Granite, and interpret segment perspectives, aggregate insights, persona-level responses, validation checks and a consultant decision brief.

What this proves

- Flexible survey/interview input: no fixed question set.
- PDF upload and extraction audit for realistic partner testing.
- Swiss public-data-grounded synthetic customer simulation.
- Customer board, use-case fit, scenario moves and individual response traceability.
- Benchmark, consistency, coverage, scenario-design and realism validation.
- Real-model proof path using IBM watsonx.ai / Granite 4 H Small, configured locally and in Code Engine.

Demo artifacts

Application:
https://visa-synthetic-research-copilot.27cqtktlikeo.eu-de.codeengine.appdomain.cloud

Repository:
https://github.com/ZihaoLu001/visa-synthetic-research-copilot

Sample PDF:
Federal Reserve mobile-payments survey excerpt adapted for Visa demo use.

Live model:
IBM watsonx.ai in eu-de, default model ibm/granite-4-h-small. Code Engine health reports watsonx_configured=true.

Public-data prototype. No Visa internal or client-sensitive data is used.

START HERE

Reviewer Workflow At A Glance

VISA

Who should use this?

Visa Consulting & Analytics reviewers, IBM mentors, or student evaluators who want to test the prototype with a survey/interview artifact instead of a hardcoded prompt.

12
Default live Granite proof respondents

96
Full presentation panel available

PDF
Recommended partner upload format

C
Latest watsonx example evidence grade

POSITIONING

This is an early-stage synthetic customer lab. It helps consultants identify promising propositions, weak assumptions, message risks and real-research priorities before commissioning real customer research.

Four-step operation

1 UPLOAD
Attach a PDF survey/interview guide or use the sample public survey.

2 REVIEW
Check extracted scenario text and edit questions if needed.

3 CONFIGURE
Set target market and adjust card propositions.

4 RUN
Generate synthetic customer lens, insights, persona responses and validation evidence.

RECOMMENDED REVIEWER SETTING

Use **Model provider = watsonx** for the final real-model proof. Keep **mock** only as a fallback for rehearsal or quota issues. Use **Quick real-model proof** for a fast live run, then move to **Full survey** and **96 respondents** when quota/time allows.

NOT A TOY DEMO

Model, Algorithm And Delivery Stack

VISA

Real LLM path

IBM watsonx.ai is the primary provider. The app calls Granite through the ibm-watsonx-ai ModelInference abstraction, defaulting to ibm/granite-4-h-small in eu-de.

Fallback path

MockLLM is deterministic and exists only for CI, rehearsal and classroom quota contingency. It is not presented as real customer intelligence.

Core algorithm

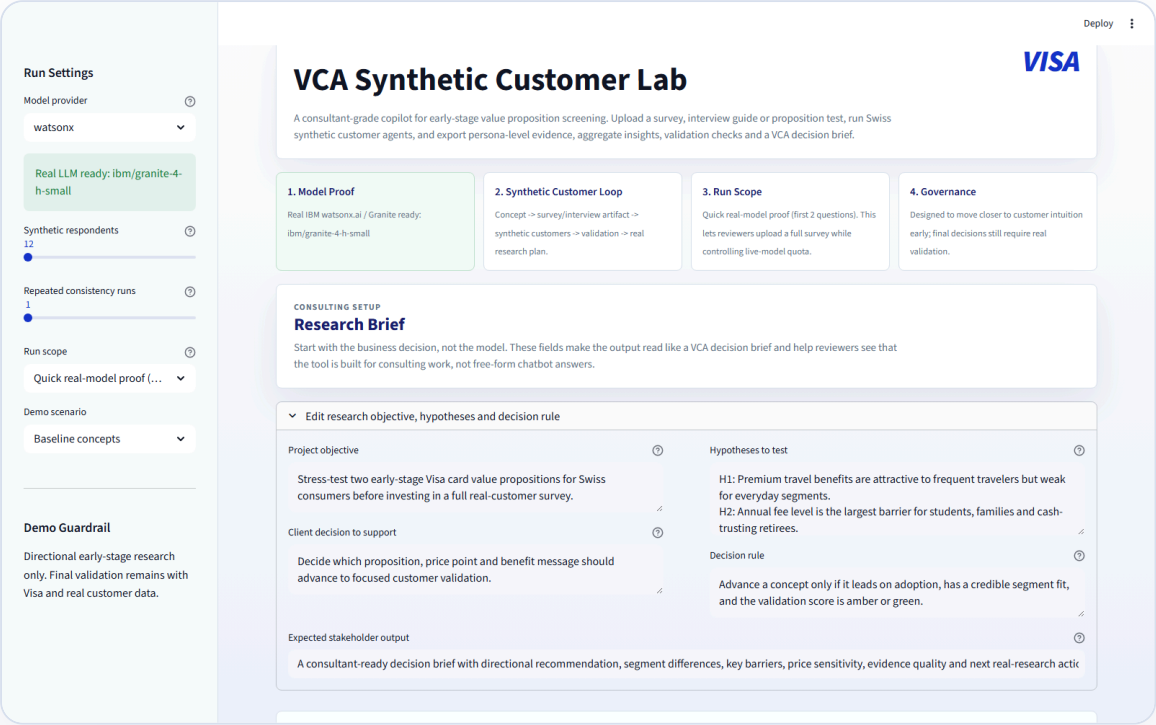
Extract survey/interview text, parse structured questions, normalize key constructs, sample Swiss micro-personas, run one synthetic customer agent per persona, aggregate weighted results, validate, and synthesize a VCA decision brief.

Synthetic customer lens

The survey is the input artifact; the product output is a synthetic customer board with use-case fit, need states, objections, message tests, scenario moves, decision drivers, time/cost advantage and real-customer bridge.

Consultant quality layer

The app adds evidence grade, decision risk, lead-margin interpretation, risk flags, survey repair plan and real-customer validation plan so the output is actionable rather than decorative.



Consultant deliverable

Each run exports a polished PDF report and a delivery pack with a decision brief, synthetic customer lens JSON, quality layer JSON, consultant report, persona CSV, validation JSON, full run JSON, source audit and governance notes.

Consulting guardrail

Synthetic results are directional. They help screen concepts and design better real customer research; they do not replace final Visa validation.

INPUT ARTIFACT

Step 1 - Upload A Real Survey PDF

Preferred format

Use PDF for partner review because it matches how survey decks and interview guides are usually shared.

Sample file

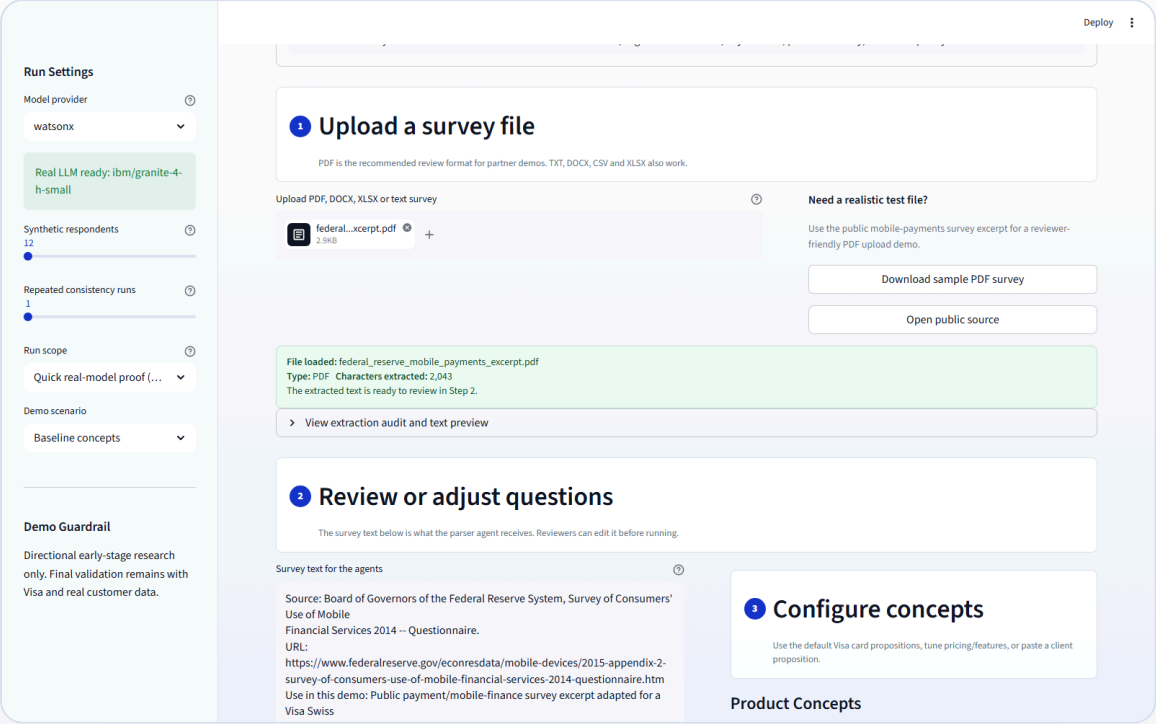
The right panel lets reviewers download the included public PDF survey excerpt and upload it immediately.

Extraction proof

After upload, the app shows file name, file type and extracted character count before any agent run.

Data guardrail

The sample comes from a public Federal Reserve questionnaire; no Visa internal data is required.



Public source: https://www.federalreserve.gov/econresdata/mobile-devices/2015-appendix-2-survey-of-consumers-use-of-mobile-financial-services-2014-questionnaire.htm

SURVEY PARSER INPUT

Step 2 - Review Or Adjust Questions

Editable extracted text

The textarea is the exact survey text sent to the parser agent. Reviewers can leave it unchanged or edit wording.

Why this matters

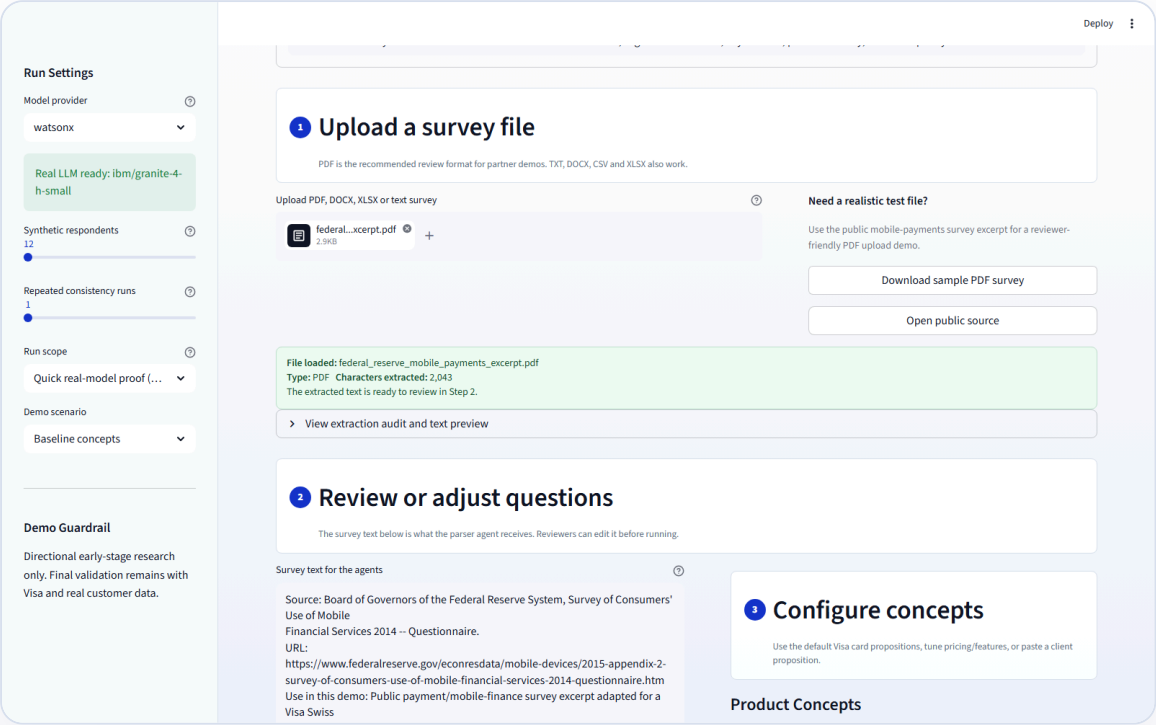
This directly addresses the Visa clarification that the system should answer survey questions without a fixed limitation.

Supported artifacts

The same ingestion layer accepts PDF, DOCX, XLSX, CSV, TXT and Markdown.

Audit trail

If the extracted text is edited, the run records that fact in the Question Parser tab.



AGENT WORKFLOW

Step 3 - Configure Concepts And Run

Concept testing

Use the default Premium Travel Card and Everyday Cashback Card, or paste a new proposition.

Live sensitivity

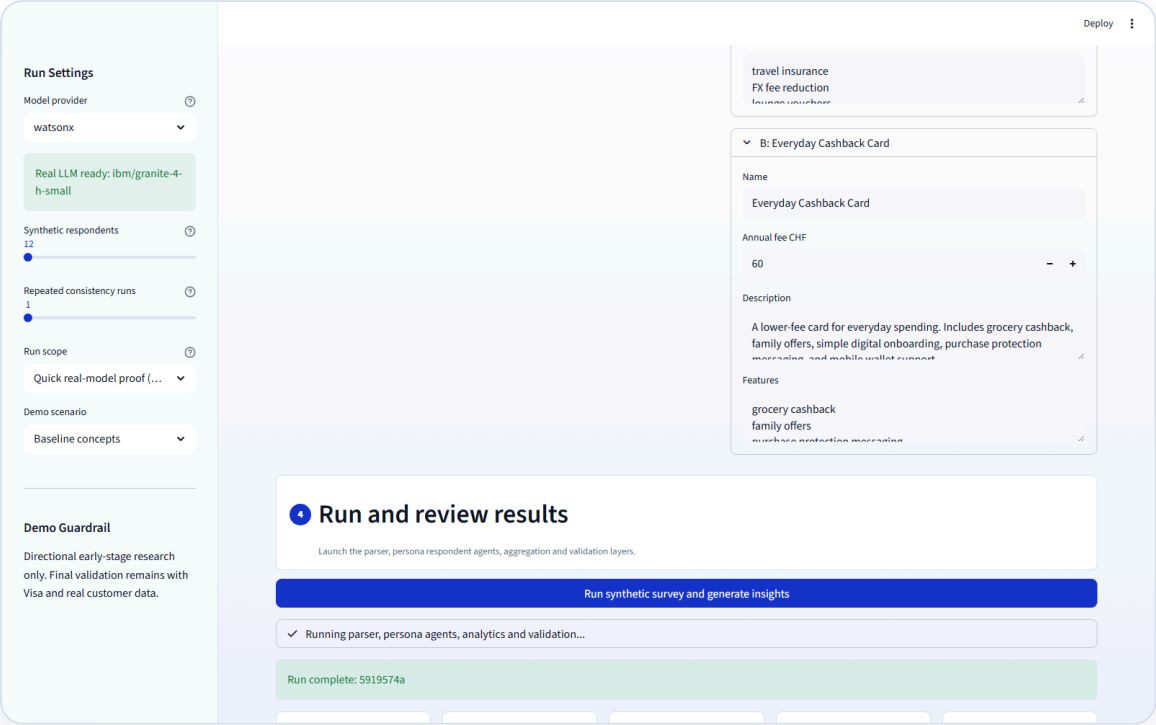
Change the annual fee or protection messaging, rerun, and compare movement in segment fit and adoption index.

Panel scale

The quick watsonx proof starts small to conserve quota. The full 96-persona run expands seven Swiss archetypes into a weighted micro-population.

Run button

The button launches survey parsing, persona respondent agents, analytics, validation and decision-brief synthesis in one workflow.



AGGREGATED OUTPUT

Step 4 - Read Consultant-Level Insights

Recommendation

The first tab summarizes the leading concept and the reason behind the result.

Adoption index

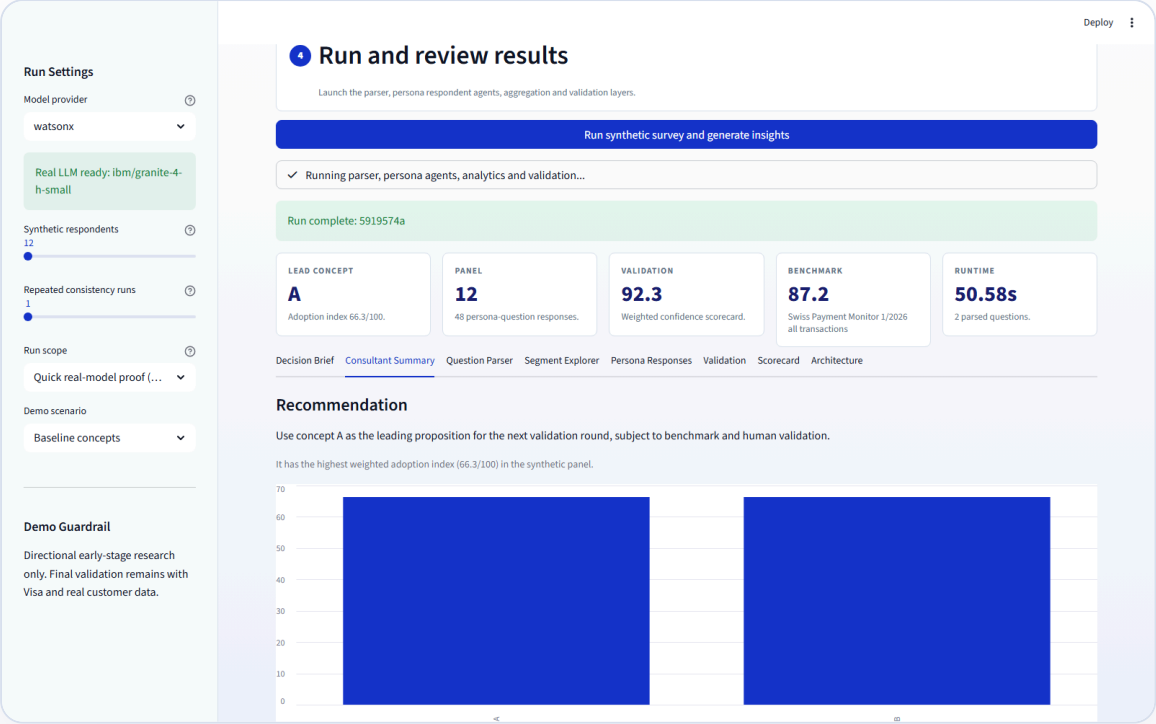
The bar chart compares concept fit across the synthetic Swiss panel.

Pricing signal

Acceptable fee ranges help frame a real customer price-sensitivity test.

Feature and barrier signals

Top labels surface recurring needs, trust drivers and objections.



QUESTION PARSER TAB

Confirm The Survey Was Parsed From PDF

Input Source Audit

Shows Source = uploaded_file, File type = pdf, character count and extraction notes.

No hardcoding

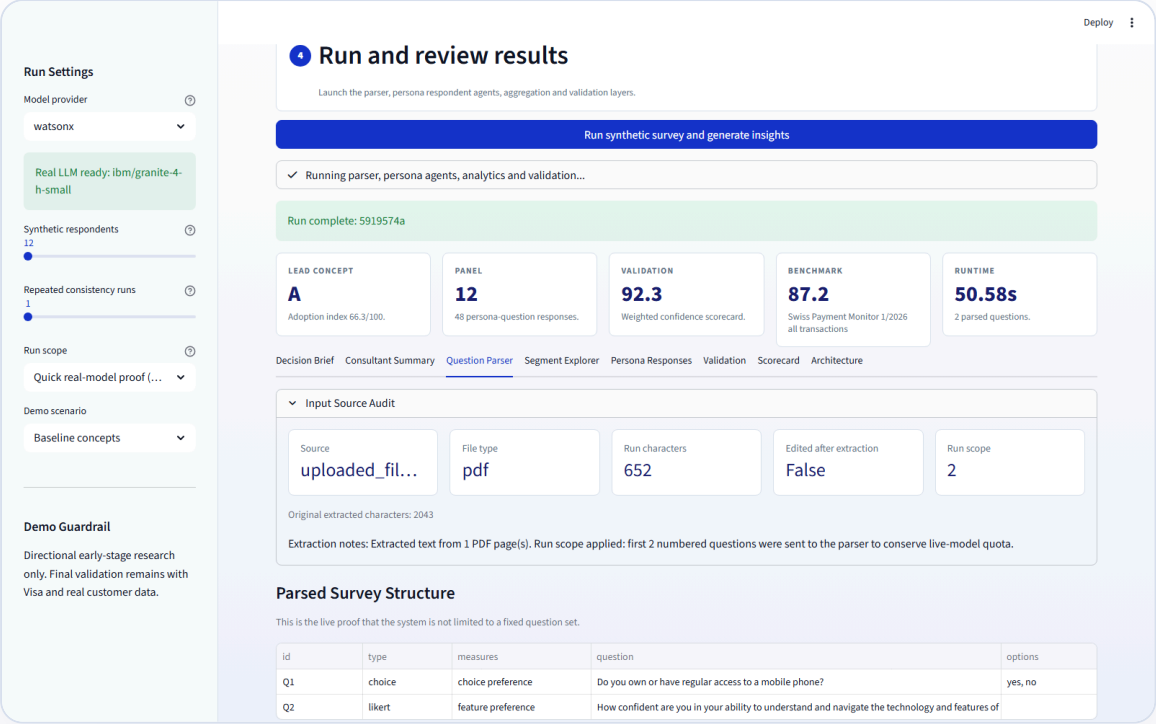
The parsed table lists question id, type, construct and options generated from the uploaded survey.

Demo expectation

Open this tab during review to prove the app can accept a new survey artifact.

Quality fix

The parser handles PDF line wrapping so wrapped questions stay together.



SEGMENT EXPLORER TAB

Inspect Segment Differences

Why segments matter

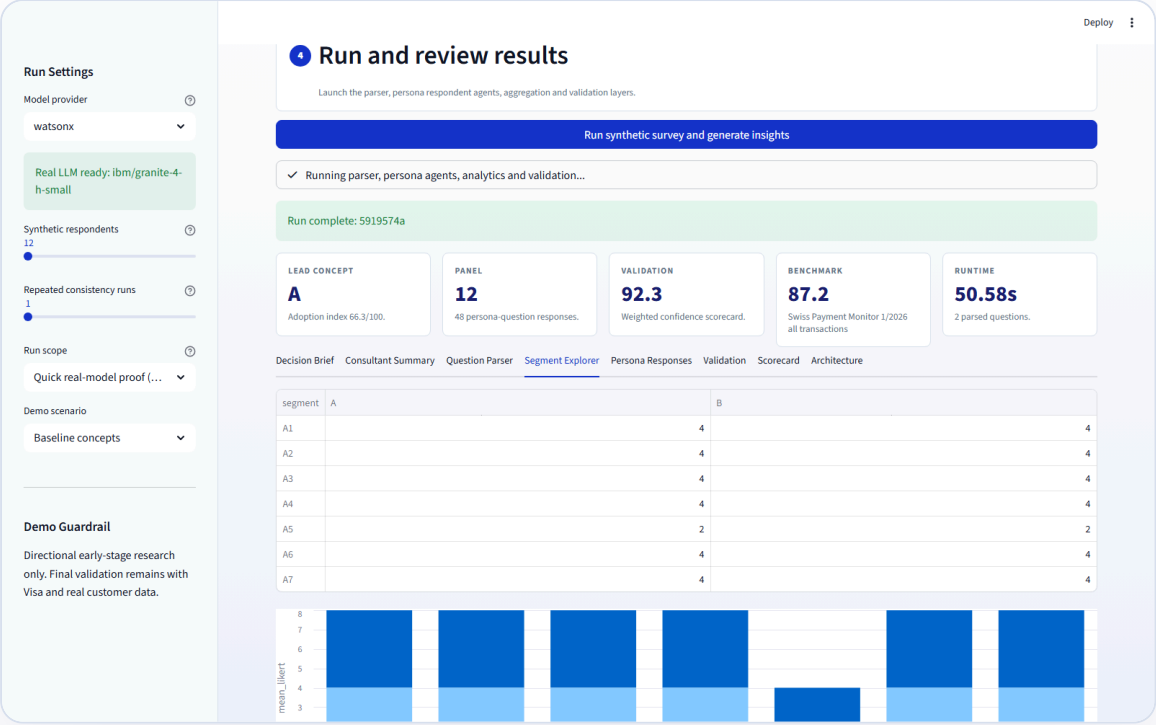
Visa asked for different personas or demographic groups, not one generic customer.

What to look for

Compare how students, family budget parents, travelers, retirees and SME owners respond differently.

Consultant use

Segment variation helps decide which customer group deserves real follow-up research.



PERSONA RESPONSES TAB

Inspect Persona-Level Traceability

Individual responses

Every row stores persona id, persona name, concept, question, answer, rationale and confidence.

Filter workflow

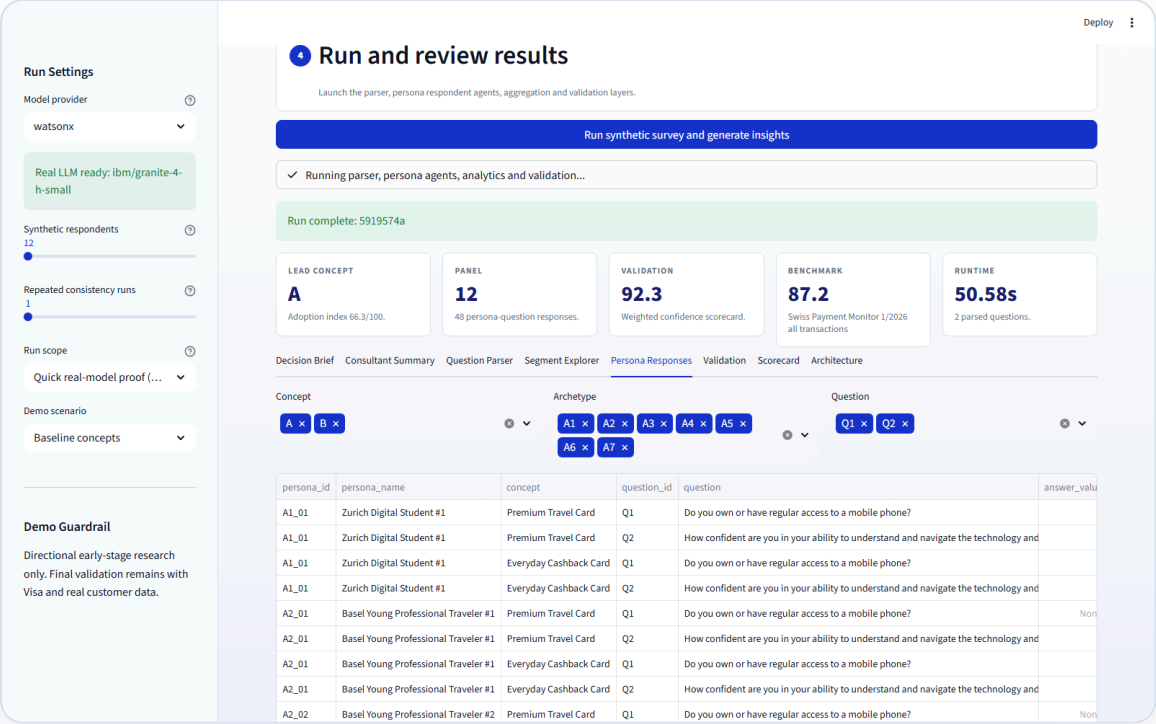
Filter by concept, archetype or question to inspect why a segment is driving the aggregate result.

Reviewer value

This satisfies the Visa request for both aggregated insights and persona-level survey responses.

Export

Download the filtered response table as CSV for analysis or appendix material.



VALIDATION TAB

Check Validation Confidence

Benchmark alignment

Compares synthetic payment behavior mix against public Swiss payment survey anchors.

Internal consistency

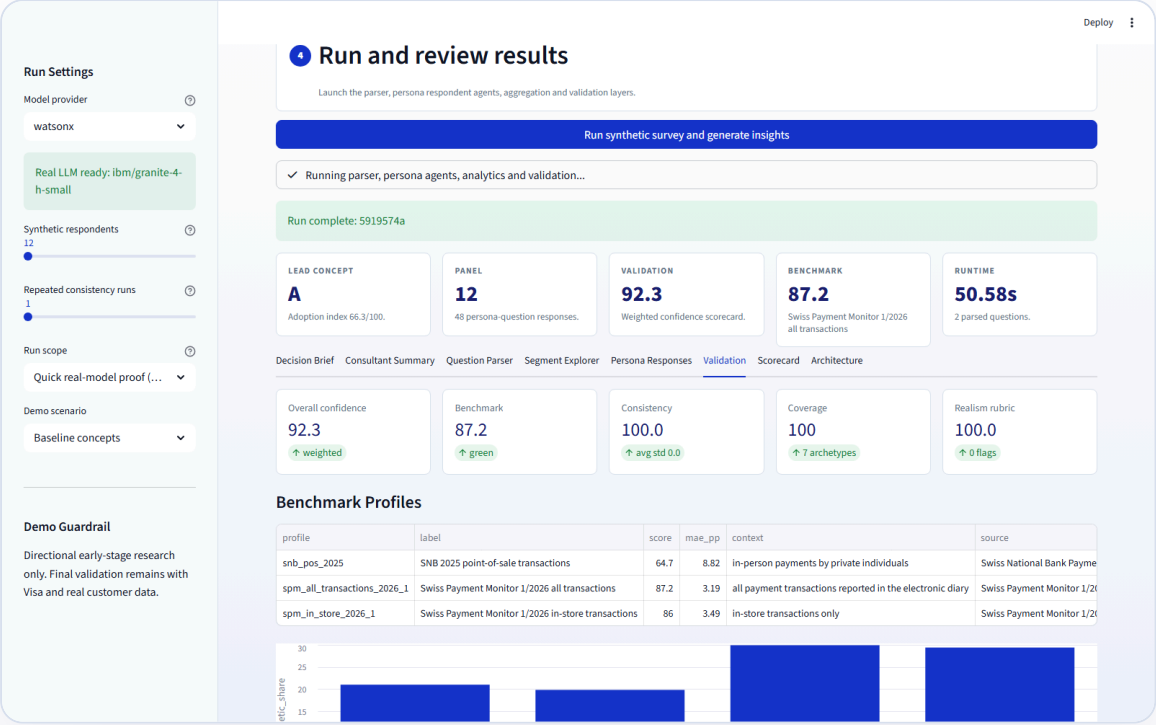
Checks repeated persona runs for response variance.

Coverage

Confirms the panel covers the Swiss archetypes and the survey covers key constructs.

Interpretation

This is directional evidence, not a claim that synthetic respondents replace real Visa validation.



DECISION BRIEF TAB

Read The Synthetic Customer Lens

Why this matters

This is the Bain/Visa framing: the output is not only completed survey rows, but a synthetic customer learning layer for proposition decisions.

Use-case fit

Shows whether the run supports value proposition design, segmentation, message testing, NPS/sentiment proxy, objection training and pricing sensitivity.

Customer board

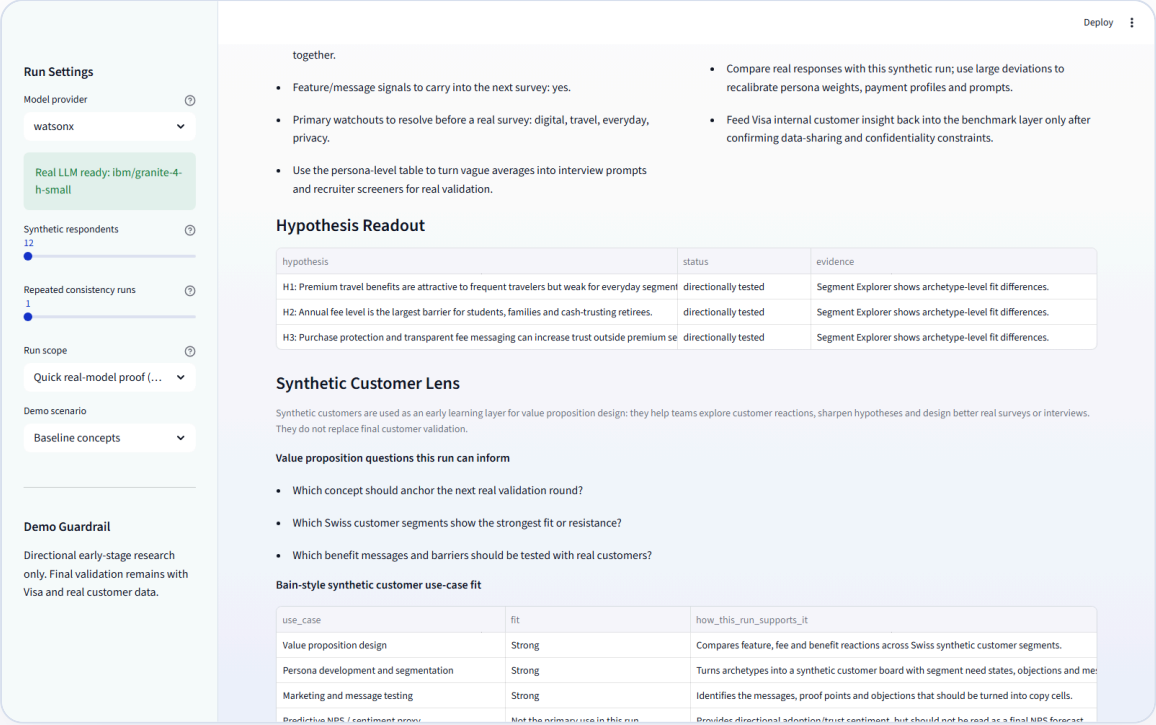
Summarizes each Swiss segment's reality, likely best-fit concept, need state, objections to probe and message to test.

Scenario moves

Turns the current run into the next experiment: price cells, message cells, segment focus and real research design.

Real-customer bridge

Keeps the governance clear: synthetic learning first, qualitative validation next, quantitative calibration after that.



Deploy

Run Settings

Model provider

watsonx

Real LLM ready: ibm/granite-4-h-small

Synthetic respondents

12

Repeated consistency runs

1

Run scope

Quick real-model proof (…

Demo scenario

Baseline concepts

Demo Guardrail

Directional early-stage research only. Final validation remains with Visa and real customer data.

together.

- Feature/message signals to carry into the next survey: yes.
- Primary watchouts to resolve before a real survey: digital, travel, everyday, privacy.
- Use the persona-level table to turn vague averages into interview prompts and recruiter screeners for real validation.
- Compare real responses with this synthetic run; use large deviations to recalibrate persona weights, payment profiles and prompts.
- Feed Visa internal customer insight back into the benchmark layer only after confirming data-sharing and confidentiality constraints.

Hypothesis Readout

hypothesis	status	evidence
H1: Premium travel benefits are attractive to frequent travelers but weak for everyday segment	directionally tested	Segment Explorer shows archetype-level fit differences.
H2: Annual fee level is the largest barrier for students, families and cash-trusting retirees.	directionally tested	Segment Explorer shows archetype-level fit differences.
H3: Purchase protection and transparent fee messaging can increase trust outside premium se	directionally tested	Segment Explorer shows archetype-level fit differences.

Synthetic Customer Lens

Synthetic customers are used as an early learning layer for value proposition design: they help teams explore customer reactions, sharpen hypotheses and design better real surveys or interviews. They do not replace final customer validation.

Value proposition questions this run can inform

- Which concept should anchor the next real validation round?
- Which Swiss customer segments show the strongest fit or resistance?
- Which benefit messages and barriers should be tested with real customers?

Bain-style synthetic customer use-case fit

use_case	fit	how_this_run_supports_it
Value proposition design	Strong	Compares feature, fee and benefit reactions across Swiss synthetic customer segments.
Persona development and segmentation	Strong	Turns archetypes into a synthetic customer board with segment need states, objections and me
Marketing and message testing	Strong	Identifies the messages, proof points and objections that should be turned into copy cells.
Predictive NPS / sentiment proxy	Not the primary use in this run	Provides directional adoption/trust sentiment, but should not be read as a final NPS forecast

DECISION BRIEF TAB

Read The Consultant Quality Layer

Evidence grade

Summarizes whether the synthetic run is strong enough for directional prioritization or still needs major repair.

Decision risk

Combines validation, lead margin and segment differentiation so consultants do not over-read a narrow synthetic lead.

Risk flags

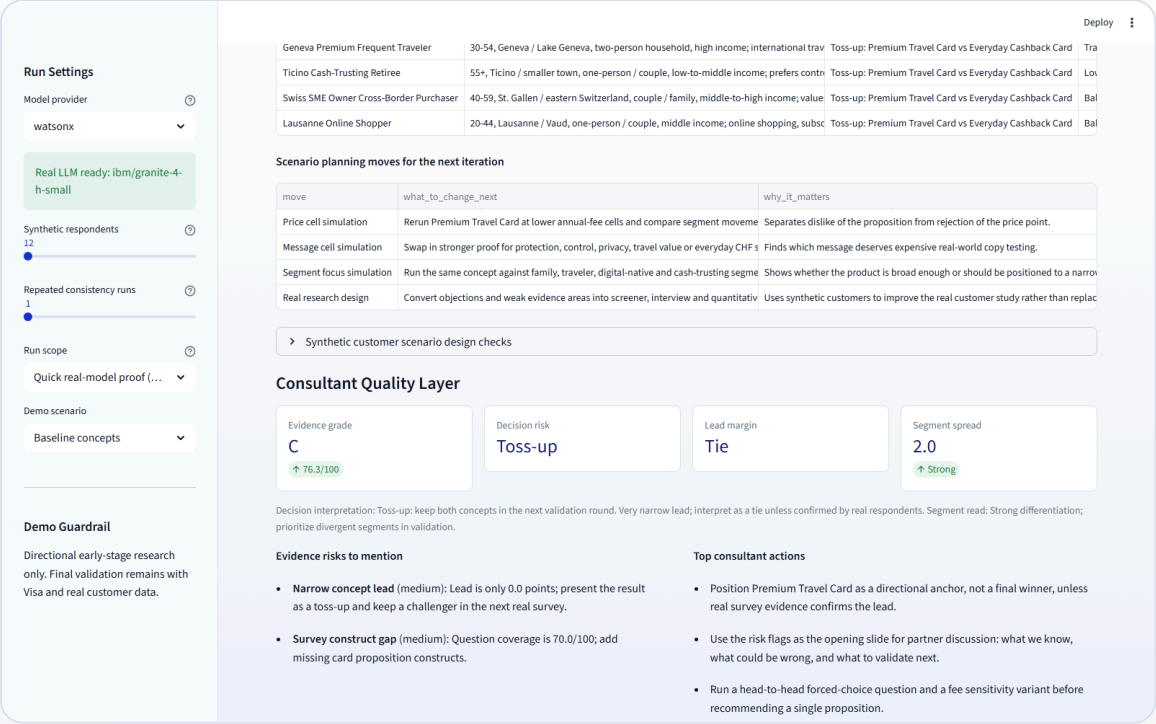
Calls out weak lead, construct gaps, realism issues or price-value mismatch in plain business language.

Survey repair plan

Recommends exact modules to add before a real Swiss customer survey, such as choice-based validation, price sensitivity, feature trade-off and barrier/trust questions.

Real validation plan

Turns the output into the next consulting action: interviews, Swiss pulse survey, pricing cell test and calibration loop.



Deploy

Run Settings

Model provider: watsonx

Real LLM ready: ibm/granite-4-h-small

Synthetic respondents: 12

Repeated consistency runs: 1

Run scope: Quick real-model proof (…

Demo scenario: Baseline concepts

Demo Guardrail

Directional early-stage research only. Final validation remains with Visa and real customer data.

Geneva Premium Frequent Traveler	30-54, Geneva / Lake Geneva, two-person household, high income; international trav	Toss-up: Premium Travel Card vs Everyday Cashback Card	Tra
Ticino Cash-Trusting Retiree	55+, Ticino / smaller town, one-person / couple, low-to-middle income; prefers contr	Toss-up: Premium Travel Card vs Everyday Cashback Card	Lo
Swiss SME Owner Cross-Border Purchaser	40-59, St. Gallen / eastern Switzerland, couple / family, middle-to-high income; value	Toss-up: Premium Travel Card vs Everyday Cashback Card	Bal
Lausanne Online Shopper	20-44, Lausanne / Vaud, one-person / couple, middle income; online shopping, subsc	Toss-up: Premium Travel Card vs Everyday Cashback Card	Bal

Scenario planning moves for the next iteration

move	what_to_change_next	why_it_matters
Price cell simulation	Rerun Premium Travel Card at lower annual-fee cells and compare segment moveme	Separates dislike of the proposition from rejection of the price point.
Message cell simulation	Swap in stronger proof for protection, control, privacy, travel value or everyday CHF s	Finds which message deserves expensive real-world copy testing.
Segment focus simulation	Run the same concept against family, traveler, digital-native and cash-trusting segme	Shows whether the product is broad enough or should be positioned to a narro
Real research design	Convert objections and weak evidence areas into screener, interview and quantitativ	Uses synthetic customers to improve the real customer study rather than replac

Synthetic customer scenario design checks

Consultant Quality Layer

Evidence grade	Decision risk	Lead margin	Segment spread
C (76.3/100)	Toss-up	Tie	2.0 (Strong)

Decision interpretation: Toss-up: keep both concepts in the next validation round. Very narrow lead; interpret as a tie unless confirmed by real respondents. Segment read: Strong differentiation; prioritize divergent segments in validation.

Evidence risks to mention

- **Narrow concept lead** (medium): Lead is only 0.0 points; present the result as a toss-up and keep a challenger in the next real survey.
- **Survey construct gap** (medium): Question coverage is 70.0/100; add missing card proposition constructs.

Top consultant actions

- Position Premium Travel Card as a directional anchor, not a final winner, unless real survey evidence confirms the lead.
- Use the risk flags as the opening slide for partner discussion: what we know, what could be wrong, and what to validate next.
- Run a head-to-head forced-choice question and a fee sensitivity variant before recommending a single proposition.

EVALUATION MAPPING

Use The Scorecard In The Final Presentation

Pilot readiness gate

The top of the Scorecard tab shows which checks are ready for partner review and which need attention.

Real model status

For final proof, Evidence mode should show Real IBM watsonx.ai and ibm/granite-4-h-small.

Demo

Running app, PDF upload, parser, agents, results and validation are all live.

Architecture

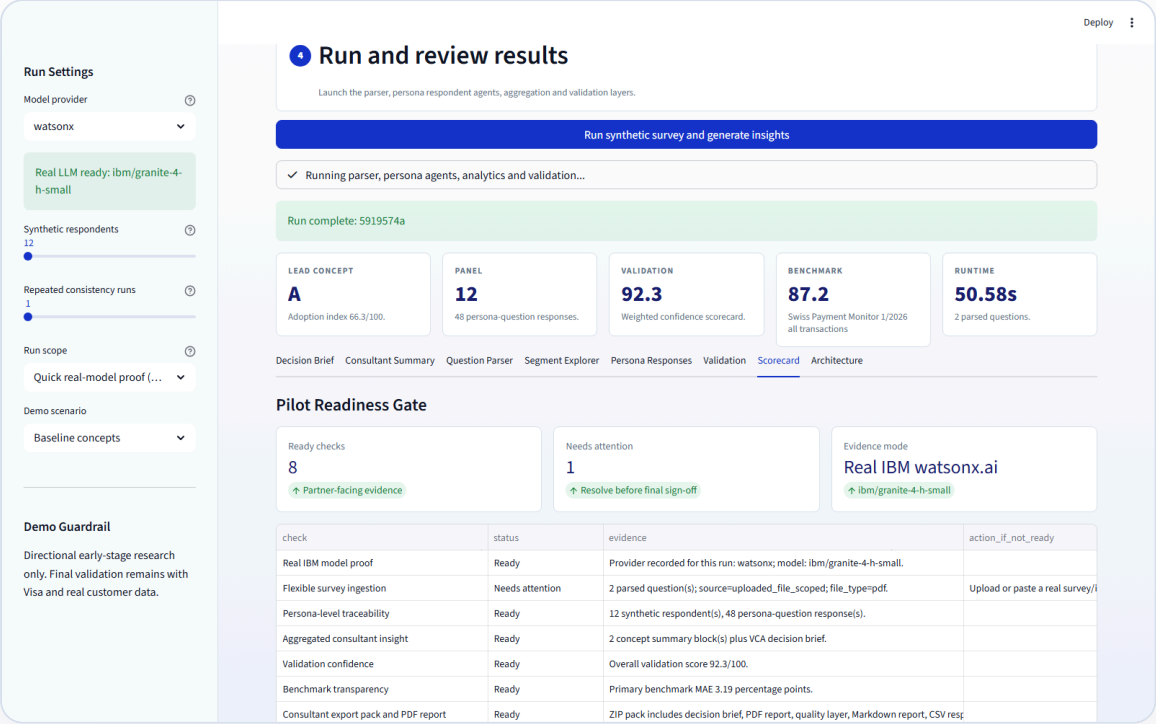
UI, ingestion, parser, persona store, orchestrator, respondent agents, analytics and validator are visible.

KPIs

Runtime, response count, parse success, consistency and benchmark scores are quantified.

Business value

Position the system as early-stage concept screening and better real-survey design.



PORTABLE PARTNER ARTIFACT

Export PDF Report And Delivery Pack

Why this matters

VCA reviewers should not need to trust only the live screen. They can download a consultant-style PDF report and inspect the evidence offline.

PDF report

The report includes executive answer, Synthetic Customer Lens, use-case fit, scenario checks, customer board, consultant quality layer, decision matrix, segment fit, persona evidence, validation confidence, methodology and limitations.

Included files

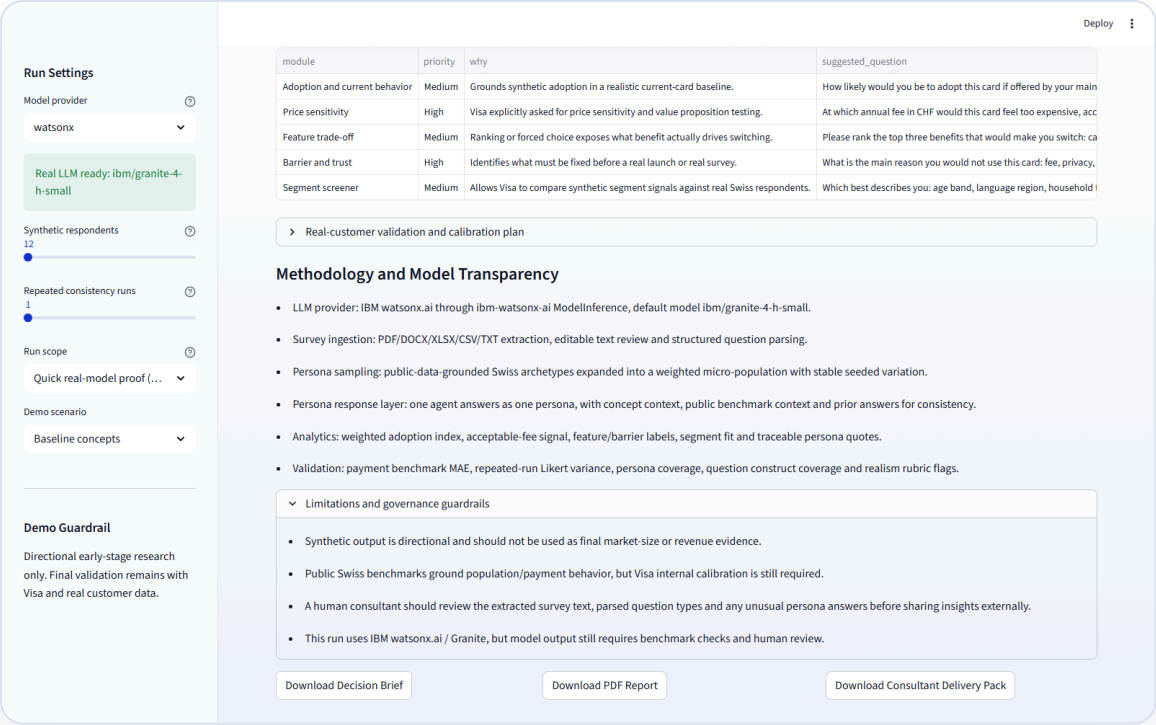
The ZIP contains the PDF report, decision brief, synthetic customer lens JSON, quality layer JSON, consultant report, persona-level CSV, validation JSON, full run JSON, input-source audit, methodology/governance notes and pilot-readiness gate.

Review workflow

Send the input PDF survey plus output PDF report together with the app URL so IBM/Visa can check recommendation logic, persona traceability and validation evidence.

No secrets

The delivery pack contains run evidence, not watsonx API keys or local credentials.



PARTNER ALIGNMENT

Suggested Slack Message To Visa / IBM

Message draft

Hi Visa / IBM team,

We prepared a short operation manual and a sample input/output pair for the current prototype. It shows a real PDF survey/interview artifact upload, dynamic parsing, IBM watsonx/Granite model path, Swiss synthetic customer responses, a segment customer board, Bain-style use-case fit, aggregated consultant insights, a Consultant Quality Layer, validation checks, and a generated consultant PDF report.

Attached are the input PDF survey and the generated output PDF report from the same flow.

Could you please let us know whether this matches your expected direction for the final Visa use case?

1. Is the Synthetic Customer Lab framing aligned with your intended use case?
2. Is the flexible PDF survey/interview upload flow aligned with your expectation as the input artifact?
3. Should the final output emphasize aggregate insights, persona-level responses, synthetic customer segment cards, or all three?
4. Is the evidence grade / decision risk / survey repair layer useful for VCA consultants?
5. Are the benchmark, consistency, coverage and realism validation checks useful enough for VCA consultants?
6. Is there any missing use-case requirement we should address before finals?

Thank you!

What to emphasize verbally

- **Flexible input:** The system is not a fixed Q&A bot.
- **Public-data grounding:** Personas are used because internal Visa customer data is unavailable.
- **Consultant value:** It produces early directional evidence, a decision posture, evidence quality and better real survey design.
- **Validation discipline:** The app exposes limitations through benchmark and consistency checks.

BEST CHANNEL

Use the Visa-specific Slack channel for use-case/product feedback. Use the general guest channel for deployment or IBM platform issues.